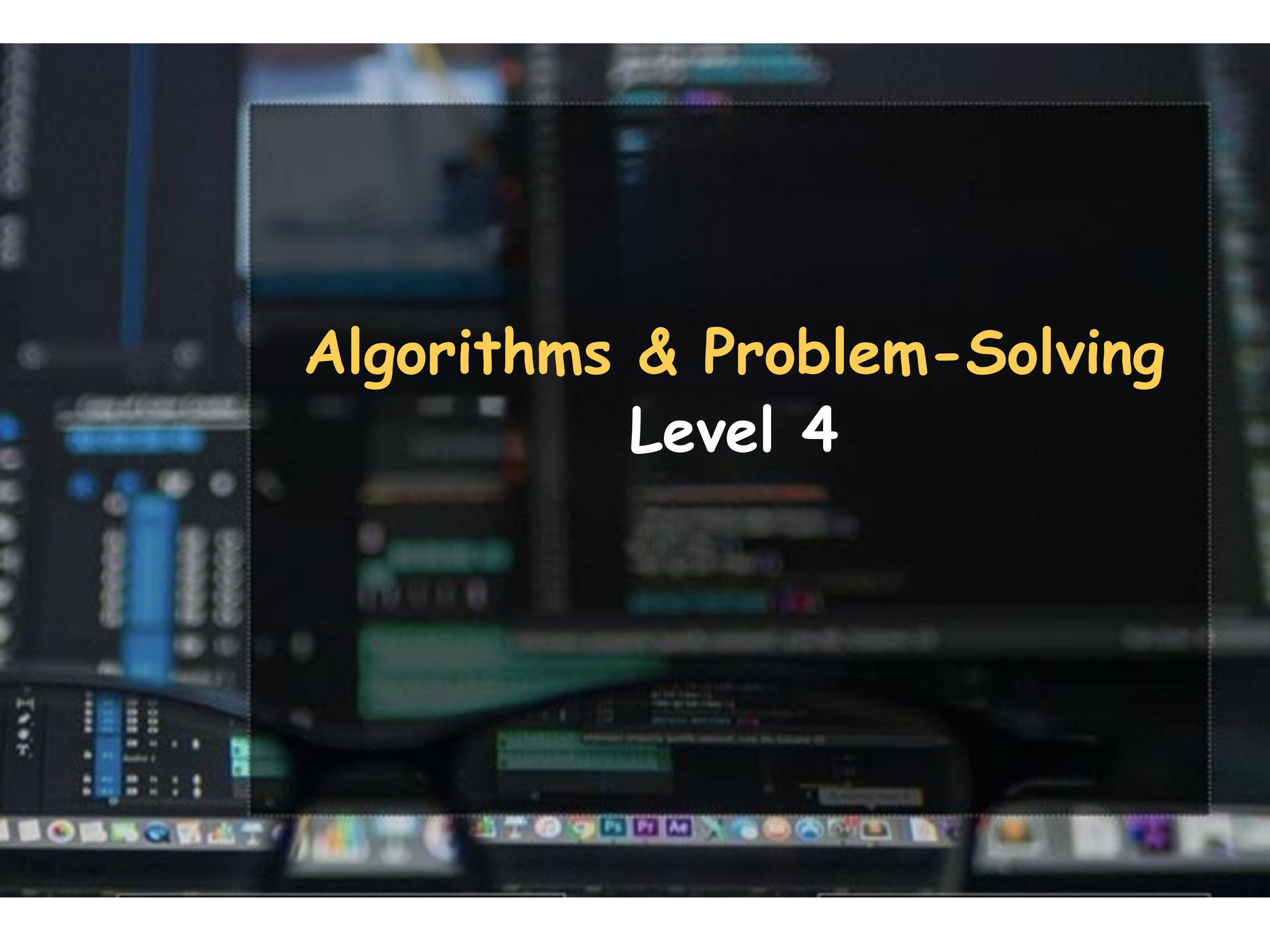
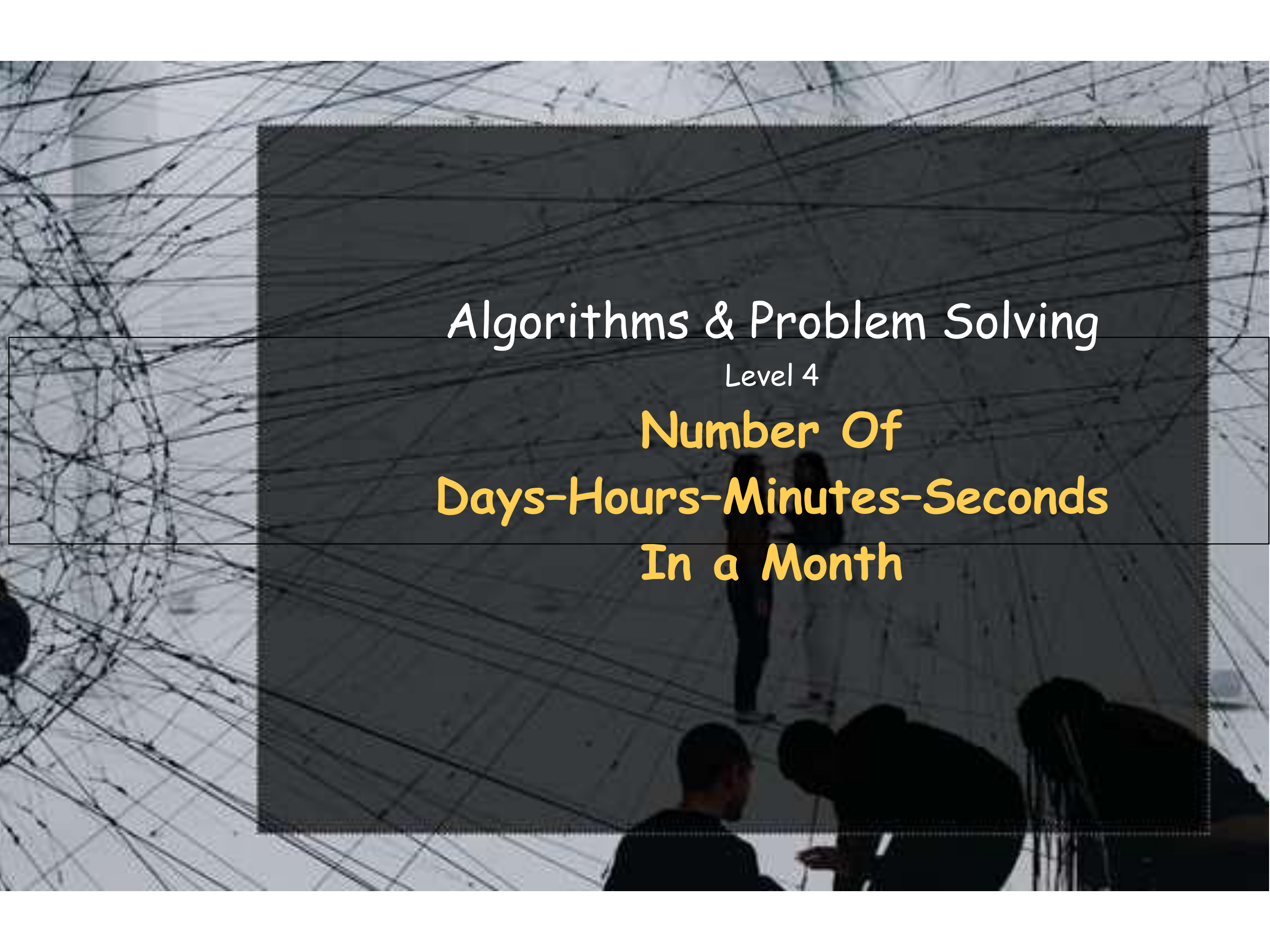




Algorithms & Problem-Solving Level 4

The background of the slide features a complex, abstract pattern of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some appearing to be in a group, which adds a human element to the technical or network-like background.

Algorithms & Problem Solving

Level 4

Number Of Days-Hours-Minutes-Seconds In a Month



Definition

- The goal is to create a program to calculate the number of days, hours, minutes, and seconds in a specific month of a specific year, as defined by the user.



Explanation

- The program asks the user to:
 - ① Enter the year
 - ② Enter the month number
- Then it calculates the number of days in that month (taking into account whether it is a leap year or not), and then calculates:
 - $\text{Number of hours} = \text{Days} \times 24$
 - $\text{Number of minutes} = \text{Hours} \times 60$
 - $\text{Number of seconds} = \text{Minutes} \times 60$

Procedures

- 1 The ``readYear()`` function
 -  Requires the user to enter the year and returns it as a value.
- 2 The ``isLeapYear(int year)`` function
 -  Checks if the year is a leap year.
 -  A leap year is divisible by 4 but not by 100, unless it is divisible by 400.
- 3 The ``readMonth()`` function
 -  Requires the user to enter the month number.
- 4 The ``getDaysInMonth(int month, int year)`` function
 -  Specifies the number of days in a given month.
 -  Months 1, 3, 5, 7, 8, 10, 12 → 31 days
 -  Months 4, 6, 9, 11 → 30 days
 -  Month 2 → 28 or 29 days depending on whether it's a leap year.
- 5 The ``printMonthDetails()`` function
 -  The function calculates and displays the days, hours, minutes, and seconds to the user.
- 6 The `main()` function
 -  executes all the previous steps sequentially.



Example:

Please enter a year to check? 2000

Please enter a Month to check? 2

Number of Days in Month [2] is 29

Number of Hours in Month [2] is 696

Number of Minutes in Month [2] is 41760

Number of Seconds in Month [2] is 2505600



Note:

■ If the user enters an invalid month number (greater than 12 or less than 1), the function will return the value 0

Summary

- ✓ The program applies the concepts of:
 - Functions
 - Conditions
 - Arithmetic operations
 - Using switch-case

Thank you 😊